

OBLUCHYE, Russian Federation

WMO: 317020

Lat: 49.0250N

Lon: 131.0493E

Elev: 877

StdP: 14.24

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.1	-29.6	-39.1	0.6	-32.8	-35.6	0.7	-29.4	10.7	-7.6	9.6	-10.1	3.0	0	0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	84.9	69.9	81.7	68.5	78.9	67.0	72.6	80.6	70.9	78.8	69.2	76.3	4.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.1	114.6	76.9	68.2	107.1	75.2	66.5	100.9	73.3	37.0	80.7	35.4	79.2	34.0	76.4	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.8	10.0	8.8	-38.2	90.9	3.8	2.7	-41.0	92.8	-43.2	94.4	-45.4	95.9	-48.2	97.8
			-38.5	76.3	3.7	2.8	-41.1	78.3	-43.3	80.0	-45.4	81.6	-48.1	83.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.7	-11.3	-2.6	15.4	37.0	51.8	62.6	68.7	65.1	53.2	35.6	10.9	-8.4
	DBStd	29.68	8.14	9.59	11.67	8.56	6.83	6.13	4.40	5.30	6.98	8.88	12.82	9.04
	HDD50	8392	1901	1473	1073	399	60	1	0	1	46	453	1174	1811
	HDD65	12412	2366	1893	1538	841	411	115	15	64	356	913	1624	2276
	CDD50	1702	0	0	0	8	117	379	580	469	143	6	0	0
	CDD65	246	0	0	0	0	2	43	131	67	3	0	0	0
	CDH74	2092	0	0	0	12	96	512	957	483	32	0	0	0
	CDH80	473	0	0	0	1	14	140	242	74	2	0	0	0
Wind	WSAvg	4.1	3.4	3.9	4.3	4.8	5.0	4.3	3.9	3.7	4.0	4.2	3.9	3.5
Precipitation	PrecAvg	31.1	0.4	0.4	1.0	1.7	3.3	4.4	6.3	6.1	3.5	2.0	1.2	0.7
	PrecMax	41.9	1.4	1.4	2.8	5.2	7.4	8.0	9.5	10.0	6.7	4.0	3.6	2.5
	PrecMin	24.4	0.0	0.0	0.0	0.3	0.9	1.3	2.0	2.5	1.1	0.5	0.1	0.2
	PrecStd	4.4	0.3	0.4	0.7	1.0	1.5	1.9	1.8	2.0	1.3	0.9	0.7	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.8	29.0	48.1	74.3	81.6	89.6	89.7	85.2	78.3	67.0	44.3	20.8
		MCWB	14.3	24.0	37.1	51.1	57.9	68.9	71.2	72.2	65.0	51.8	36.4	18.7
	2%	DB	10.0	23.3	42.5	63.7	76.7	83.6	85.9	82.0	73.7	59.4	37.6	14.6
		MCWB	8.1	19.6	33.6	46.5	56.2	66.7	71.3	70.6	62.0	47.3	31.9	12.9
	5%	DB	6.0	18.6	37.7	57.6	71.8	79.7	82.4	79.1	70.1	54.1	33.1	11.1
		MCWB	4.5	15.7	30.7	43.5	54.4	65.2	70.5	68.7	59.7	43.9	30.1	9.7
	10%	DB	2.4	13.7	33.5	52.1	67.3	75.9	79.4	76.3	66.1	49.6	29.5	6.3
		MCWB	1.2	11.2	28.4	40.7	52.3	63.6	69.0	67.3	58.3	42.0	26.7	5.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.2	24.4	38.0	52.2	61.9	71.8	75.7	74.7	68.0	53.6	37.1	19.0
		MCDB	16.5	28.9	47.2	69.9	75.2	86.3	84.6	81.3	74.1	63.2	43.1	20.5
	2%	WB	8.1	19.8	34.2	47.9	58.8	68.7	73.0	72.4	64.4	49.1	32.8	13.2
		MCDB	9.8	23.1	40.9	61.6	70.9	79.4	81.4	79.2	70.1	56.4	36.1	14.6
	5%	WB	4.6	15.8	31.5	44.3	56.4	66.7	71.5	70.5	61.9	45.5	30.5	9.7
		MCDB	6.0	18.6	36.7	56.2	67.9	77.1	79.9	77.4	68.1	52.5	33.8	11.1
	10%	WB	1.3	11.3	28.7	41.1	54.0	64.8	70.0	68.4	59.4	42.6	26.6	5.2
		MCDB	2.4	13.6	33.6	50.9	65.5	74.3	77.4	74.3	64.6	48.8	29.5	6.4
Mean Daily Temperature Range	5% DB	MDBR	22.1	25.4	24.1	21.6	22.7	20.2	17.8	18.5	21.2	20.1	19.4	20.0
		MCDBR	24.3	28.1	24.6	31.5	32.5	28.3	24.5	23.3	27.8	28.7	19.8	21.4
		MCWBR	22.5	24.3	16.4	17.0	15.6	13.2	12.1	12.4	15.8	16.8	14.7	20.0
	5% WB	MCDBR	24.3	27.4	22.2	29.0	26.8	23.7	20.9	19.9	22.7	25.3	18.3	21.3
		MCWBR	22.5	23.7	14.7	16.2	14.4	12.1	11.5	11.5	14.7	15.9	13.8	19.8
Clear-Sky Solar Irradiance	taub	0.286	0.319	0.396	0.537	0.584	0.520	0.498	0.446	0.413	0.427	0.354	0.292	
	taud	2.133	2.024	1.871	1.711	1.701	1.947	2.062	2.204	2.199	1.989	2.060	2.101	
	Ebn at Noon	237	254	250	225	221	239	243	251	246	209	201	213	
	Edh at Noon	26	37	52	69	72	57	50	42	37	39	29	24	
All-Sky Solar Radiation	RadAvg	579	970	1432	1532	1634	1745	1627	1483	1206	805	540	449	
	RadStd	16	22	87	160	156	234	157	108	121	60	40	21	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page